

Learning Objective 1.4: I can calculate input impedance, feed considerations, and the role of baluns in an antenna feed system.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Radiation resistance and efficiency.** A short dipole has length $\ell = 0.08\lambda$. Its conductors contribute a loss resistance of $R_{\text{loss}} = 1.5\Omega$.

(a) Compute the radiation resistance $R_{\text{rad}} = 80\pi^2(\ell/\lambda)^2$.

$R_{\text{rad}} =$

(b) Compute the radiation efficiency η_{rad} .

$\eta_{\text{rad}} =$

(c) The short dipole's directivity is $D = 1.5$ (1.76 dBi).
Find the gain in dBi.

$G =$

2. Feeding a half-wave dipole with 50Ω coax.

- (a) A dipole trimmed to resonance presents $Z_{\text{in}} = 70 + j0\Omega$. Find Γ , VSWR, return loss, and mismatch loss on a 50Ω line.

VSWR =

RL =

- (b) An *untrimmed* half-wave dipole presents $Z_{\text{in}} = 73 + j42.5\Omega$. Find $|\Gamma|$ and VSWR on the same line.

 $|\Gamma|$ =

VSWR =

- (c) Compare (a) and (b): in one sentence, what did trimming to resonance buy you, and why?

3. **Quarter-wave transformer.** A resonant antenna presents a real input impedance $R_L = 36\Omega$. You want to match it to a 50Ω feed line with a quarter-wave transformer.

- (a) What characteristic impedance Z_1 must the $\lambda/4$ section have?

$Z_1 =$

- (b) What is Γ at the design frequency?

- (c) In one sentence, what happens to the match off the design frequency, and why?

4. **VSWR, reflection, and the impedance behind it.** An antenna measures $\text{VSWR} = 1.5$ on a 50Ω line.

(a) Find $|\Gamma|$.

$$|\Gamma| = \boxed{}$$

(b) Find the return loss in dB.

$$\text{RL} = \boxed{}$$

(c) Find the mismatch loss in dB.

$$L_{\text{mm}} = \boxed{}$$

(d) Assuming a purely resistive antenna, what two real Z_{in} are consistent with this VSWR?

$$Z_{\text{in}} = \boxed{}$$

5. **Mismatch that kills the amplifier.** A power amplifier (PA) delivers a forward power of $P_{\text{fwd}} = +40$ dBm toward a transmit antenna. A shunt protection diode at the PA output fails catastrophically if it must dissipate more than $P_{\text{max}} = +30$ dBm. Model *all* of the reflected power as landing on the diode.

(a) Convert P_{fwd} and P_{max} to watts.

(b) In terms of $|\Gamma|$ at the antenna, how much power is reflected back toward the PA?

(c) Find the largest $|\Gamma|$ the system can tolerate before the diode is destroyed.

(d) Convert to a VSWR limit and a return-loss limit, and state the datasheet rule.

(e) In one sentence: how does a circulator or isolator between the PA and antenna change this picture?

6. Baluns.

- (a) In one or two sentences, explain why feeding a dipole *directly* with coax (no balun) distorts the pattern.

- (b) You must feed a folded dipole ($Z_{\text{in}} \approx 292\Omega$) with 75Ω coax. Which balun would you choose, and what does it do?

- (c) You feed a straight resonant dipole ($\approx 70\Omega$) with 50Ω coax. Which balun would you choose, and why is it a *different* choice than (b)?

Documentation: _____